

Score matrix  $S$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 1 | 9 | 8 | 4 | 1 |
| 9 | 7 | 1 | 1 | 5 | 1 |
| 2 | 1 | 9 | 9 | 3 | 1 |
| 8 | 9 | 1 | 2 | 4 | 1 |

*block  
pruning*Mask  $M$ 

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 | 1 |

*compute gain*Gain matrix  $G$ 

|     |     |     |     |    |    |
|-----|-----|-----|-----|----|----|
| 0   | 0   | -30 | -28 | -4 | +1 |
| 0   | 0   | -30 | -28 | -5 | 0  |
| -30 | -30 | 0   | 0   | -7 | 0  |
| -28 | -28 | 0   | 0   | -6 | +1 |
| -4  | -5  | -7  | -6  | 0  | 0  |
| +1  | 0   | 0   | +1  | 0  | 0  |

*argmax  $G$* 

Chosen swap: columns 0 and 5

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 1 | 9 | 8 | 4 | 1 |
| 9 | 7 | 1 | 1 | 5 | 1 |
| 2 | 1 | 9 | 9 | 3 | 1 |
| 8 | 9 | 1 | 2 | 4 | 1 |

*permute  
columns*

After permutation

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 1 | 9 | 8 | 4 | 1 |
| 1 | 7 | 1 | 1 | 5 | 9 |
| 1 | 1 | 9 | 9 | 3 | 2 |
| 1 | 9 | 1 | 2 | 4 | 8 |